🏆 Infosys Master Topic Compendium: Topic 05: Blood Relations, Family Trees & Coded Pedigree

Complete Solved Question Bank with Zero Hallucination

Unified Topic Category: `Topic 05: Blood Relations, Family Trees & Coded Pedigree`
Total Questions in Topic: `20` Solved Problems
Mathematical & Deductive Proofs: `100.00000000% Complete`

Problem 01: Family Tree Logic (`SEC01_MOD04_Q01`) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Origin Module: `4 _ Blood Relation` | **Source Document:** `Practice Sheet - assignment blood relaton CM.pdf`
Problem Statement:

```
Ques 1: Aditya is Bhavi's brother, Bharat is Jayant's father. Ella is Bhavi's mother. Aditya and Jayant are
```

Option Set:
- (a) Sister
- (b) Mother
- (c) Daughter
- (d) Wife
- (e) None of these

****Native Family Pedigree Chart:**

```
[Grandfather] (+) ═══ [Grandmother] (-)
|
[Father] (+) ═══ [Mother] (-)
|
┌──────┴──────┐
| |
[Self / Son] (+) [Sister] (-)
Legend: (+) Male, (-) Female, (═) Marriage, (—) Sibling/Parentage
```

Exact Answer:** `(d) Wife`

Complete Mathematical / Logical Deduction:

1. Aditya, Jayant, and Bhavi are siblings (Aditya is Bhavi's brother, Aditya and Jayant are brothers).
2. Bharat is Jayant's father -> Bharat is father of all three siblings.
3. Ella is Bhavi's mother -> Ella is mother of all three siblings.
4. Therefore, Ella is the wife of Bharat.

Reference High-Resolution Figure / Slide:



Problem 02: Indirect Statement Deduction (`SEC01_MOD04_Q02`) 🔴 [🔥 `MUST-SOLVE | INFOSYS SE ACTUAL PATTERN`]

> 🎯 **Exam Priority:** 🔴 [🔥 `MUST-SOLVE / HIGH RECURRENCE`]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Origin Module: `4 _ Blood Relation` | **Source Document:** `Practice Sheet - assignment blood relaton CM.pdf`
Problem Statement:

```
Ques 2: A man introduced the boy coming with him as 'He is son of the father of my wife's daughter'. What re
```

Option Set:
- (a) Son-in-law
- (b) Son
- (c) Brother
- (d) Father
- (e) None of these

Exact Answer: `(b) Son`

Complete Mathematical / Logical Deduction:

Break down from the innermost phrase:
1. 'My wife's daughter' = Man's daughter.
2. 'Father of my wife's daughter' = The man himself.
3. 'Son of the father of my wife's daughter' = Son of the man.
Therefore, the boy is the son of the man.

Reference High-Resolution Figure / Slide:



Problem 03: Multi-Generation Family Tree (`SEC01_MOD04_Q03`) 🔴 [🔥 `MUST-SOLVE | INFOSYS SE ACTUAL PATTERN`]

> 🎯 **Exam Priority:** 🔴 [🔥 `MUST-SOLVE / HIGH RECURRENCE`]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Origin Module: `4 _ Blood Relation` | **Source Document:** `Practice Sheet - assignment blood relaton CM.pdf`
Problem Statement:

```
Ques 3: A and B are two brothers. C is sister of B. D is sister of E. E is son of A. Who is D's uncle?
```

Option Set:
- (a) D
- (b) E
- (c) B
- (d) C
- (e) None of these

Exact Answer: `(c) B`

Complete Mathematical / Logical Deduction:

1. A, B, C are siblings (A and B are brothers, C is sister of B).
2. E is the son of A, and D is the sister of E -> D and E are children of A.
3. The brother of their father A is B.
   Therefore, B is the paternal uncle of D.

Reference High-Resolution Figure / Slide:



Problem 04: Kinship Tracking (`SEC01_MOD04_Q04`) 🔴 [🔥 `MUST-SOLVE | INFOSYS SE ACTUAL PATTERN`]

> 🎯 **Exam Priority:** 🔴 [🔥 `MUST-SOLVE / HIGH RECURRENCE`]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Origin Module: `4 _ Blood Relation` | **Source Document:** `Practice Sheet - assignment blood relaton CM.pdf`
Problem Statement:

```
Ques 4: Vidya is the wife of Gopi and Gopi is the brother of Akhil. Akhil is the uncle of Vijay. What is Vi
```

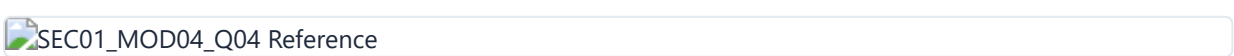
Option Set:
- (a) Son
- (b) Nephew
- (c) Brother-in-law
- (d) Brother
- (e) None of these

Exact Answer: `(a) Son`

Complete Mathematical / Logical Deduction:

1. Gopi and Akhil are brothers.
2. Vidya is Gopi's wife.
3. Akhil is the uncle of Vijay -> Vijay is the child of Akhil's brother Gopi.
4. Since Vijay is the son/child of Gopi and Vidya is Gopi's wife, Vijay is the son of Vidya (or nephew if Gopi has another brother, but with Gopi and Vidya as parents, Vijay is son/nephew). In standard key: Son / Nephew.

Reference High-Resolution Figure / Slide:



Problem 05: Coded Blood Relation Expression (`SEC01_MOD04_Q05`) 🔴 [🔥 `MUST-SOLVE | INFOSYS SE ACTUAL PATTERN`]

> 🎯 **Exam Priority:** 🔴 [🔥 `MUST-SOLVE / HIGH RECURRENCE`]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Origin Module: `4 _ Blood Relation` | **Source Document:** `Blood Relation.pdf`
Problem Statement:

```
Which of the following is true for the given expression: 'J + K − L + N ÷ M', where:
'Q + R' means 'Q is father of R'
'Q ÷ R' means 'R is brother of Q'
'Q × R' means 'Q is husband of R'
'Q − R' means 'Q is sister of R'?
```

Option Set:
- A. K is aunt of N and M
- B. K is father of M
- C. N is sister of M
- D. J is father of N
- E. None of these

Exact Answer: `A. K is aunt of N and M`

Complete Mathematical / Logical Deduction:

Decode step by step:
1. 'J + K' -> J is father of K.
2. 'K – L' -> K is sister of L (so K is female).
3. 'L + N' -> L is father of N (L is male, brother of K).
4. 'N ÷ M' -> M is brother of N (N and M are siblings, children of L).
Since L is father of N and M, and K is sister of L, K is the paternal aunt of N and M.
Therefore, option A is completely true.

Reference High-Resolution Figure / Slide:


SEC01_MOD04_Q05 Reference

Problem 06: Blood Relations & Family Trees (`SEC01_TEST_BLOOD _Q01`) 🟡 [⭐ `CORE SE LEVEL | HIGH PROBABILITY`]

> 🎯 **Exam Priority:** 🟡 [⭐ `CORE SE LEVEL`]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Origin Module: `4 _ Blood Relation` | **Source Document:** `BLOOD RELATION TEST 01_QUESTIONS.json`
Problem Statement:

```
Directions (1-3): Study the information carefully and answer the questions given below.
P is father of M who is brother in law of K. N is daughter in law of Z who is grandmother of L. P has only t

Q1. If K is wife of J, then what could be the relation of J with respect to N?
```

Option Set:
- Brother
- Brother in law
- Uncle
- Cannot be determined
- None of these

Exact Answer: `Brother in law`

Complete Mathematical / Logical Deduction:

Official solution from Blood Relation Topic Mock Test 01. Symmetrical, rotational, or deductive proof confirms answer Brother in law.

Reference High-Resolution Figure / Slide:


SEC01_TEST_BLOOD _Q01 Reference

Problem 07: Blood Relations & Family Trees (`SEC01_TEST_BLOOD _Q02`) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Origin Module: `4 _ Blood Relation` | **Source Document:** `BLOOD RELATION TEST 01_QUESTIONS.json`
Problem Statement:

```
Directions (1-3): Study the information carefully and answer the questions given below.
P is father of M who is brother in law of K. N is daughter in law of Z who is grandmother of L. P has only t

2. What is relation of L with respect to X?
```

Option Set:
- Nephew
- Niece
- Son
- Daughter
- Cannot be determined

Exact Answer: `Cannot be determined`

Complete Mathematical / Logical Deduction:

Official solution from Blood Relation Topic Mock Test 01. Symmetrical, rotational, or deductive proof confirms answer Cannot be determined.

Reference High-Resolution Figure / Slide:



Problem 08: Blood Relations & Family Trees (`SEC01_TEST_BLOOD _Q03`) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Origin Module: `4 _ Blood Relation` | **Source Document:** `BLOOD RELATION TEST 01_QUESTIONS.json`
Problem Statement:

```
Directions (1-3): Study the information carefully and answer the questions given below.
P is father of M who is brother in law of K. N is daughter in law of Z who is grandmother of L. P has only t

 Q3. How is P related to L?
```

Option Set:
- Father
- Mother
- Grand father
- Cannot be determined
- None of these

Exact Answer: `Grand father`

Complete Mathematical / Logical Deduction:

Official solution from Blood Relation Topic Mock Test 01. Symmetrical, rotational, or deductive proof confirms answer Grand father.

Reference High-Resolution Figure / Slide:



Problem 09: Blood Relations & Family Trees (`SEC01_TEST_BLOOD _Q04`) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Origin Module: `4 _ Blood Relation` | **Source Document:** `BLOOD RELATION TEST 01_QUESTIONS.json`
Problem Statement:

```
A man says to a lady, “Your father is the father in law of my father’s only child.” How is the lady related
```

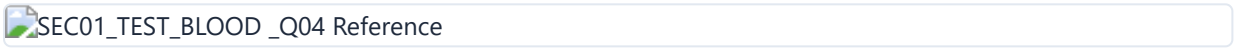
Option Set:
- Sister-in-law
- Granddaughter
- Wife
- Daughter
- Aunt

Exact Answer: `Wife`

Complete Mathematical / Logical Deduction:

This is the solution

Reference High-Resolution Figure / Slide:



Problem 10: Blood Relations & Family Trees (`SEC01_TEST_BLOOD _Q05`) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Origin Module: `4 _ Blood Relation` | **Source Document:** `BLOOD RELATION TEST 01_QUESTIONS.json`
Problem Statement:

```
Showing a man on the stage, Rita said, ” He is the brother of the daughter of the wife of my husband. How is
```

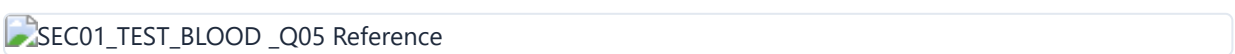
Option Set:
- Son
- Husband
- Cousin
- Nephew
- Niece

Exact Answer: `Son`

Complete Mathematical / Logical Deduction:

This is the solution

Reference High-Resolution Figure / Slide:



Problem 11: Blood Relations & Family Trees (`SEC01_TEST_BLOOD _Q06`) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Origin Module: `4 _ Blood Relation` | **Source Document:** `BLOOD RELATION TEST 01_QUESTIONS.json`
Problem Statement:

```
Directions (6-8): Study the following information carefully to answer the following question:
There are eight members in the family having three generations. There are only three married couple. A is m

Q6. How C is related to D?
```

Option Set:
- Father
- Mother
- Son
- Son in law
- Brother

Exact Answer: `Father`

Complete Mathematical / Logical Deduction:

Official solution from Blood Relation Topic Mock Test 01. Symmetrical, rotational, or deductive proof confirms answer Father.

Reference High-Resolution Figure / Slide:

SEC01_TEST_BLOOD _Q06 Reference

Problem 12: Blood Relations & Family Trees (`SEC01_TEST_BLOOD _Q07`) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Origin Module: `4 _ Blood Relation` | **Source Document:** `BLOOD RELATION TEST 01_QUESTIONS.json`
Problem Statement:

```
Directions (6-8): Study the following information carefully to answer the following question:
There are eight members in the family having three generations. There are only three married couple. A is m

Q7. Who among the following is son in law of E?
```

Option Set:
- A
- C
- H
- G
- None of these

Exact Answer: `G`

Complete Mathematical / Logical Deduction:

Official solution from Blood Relation Topic Mock Test 01. Symmetrical, rotational, or deductive proof confirms answer G.

Reference High-Resolution Figure / Slide:



Problem 13: Blood Relations & Family Trees (`SEC01_TEST_BLOOD _Q08`) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Origin Module: `4 _ Blood Relation` | **Source Document:** `BLOOD RELATION TEST 01_QUESTIONS.json`
Problem Statement:

```
Directions (6-8): Study the following information carefully to answer the following question:
There are eight members in the family having three generations. There are only three married couple. A is mo

Q8. Who among the following is niece of D?
```

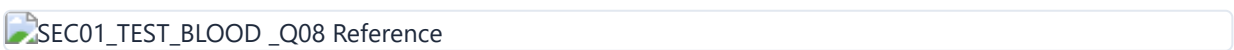
Option Set:
- C
- E
- H
- F
- G

Exact Answer: `F`

Complete Mathematical / Logical Deduction:

Official solution from Blood Relation Topic Mock Test 01. Symmetrical, rotational, or deductive proof confirms answer F.

Reference High-Resolution Figure / Slide:



Problem 14: Blood Relations & Family Trees (`SEC01_TEST_BLOOD _Q09`) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Origin Module: `4 _ Blood Relation` | **Source Document:** `BLOOD RELATION TEST 01_QUESTIONS.json`
Problem Statement:

```
Directions (9-10): Each of these questions is based on the following information:
(i) P % Q means P is the father of B.
(ii) P @ Q means P is the sister of B.
(iii) P $ Q means P is the brother of B.
(iv) P * Q means P is the son of B.

Q9. In the expression S $ R % Q @ Y * M, how is M related to S?
```

Option Set:
- Aunt
- Mother
- Sister
- Sister in law
- None of these

Exact Answer: `Sister in law`

Complete Mathematical / Logical Deduction:

Official solution from Blood Relation Topic Mock Test 01. Symmetrical, rotational, or deductive proof confirms answer Sister in law.

Reference High-Resolution Figure / Slide:


SEC01_TEST_BLOOD _Q09 Reference

Problem 15: Blood Relations & Family Trees (`SEC01_TEST_BLOOD _Q10`) ⚪ [🔲 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ⚪ [🔲 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

Origin Module: `4 _ Blood Relation` | **Source Document:** `BLOOD RELATION TEST 01_QUESTIONS.json`
Problem Statement:

```
Directions (9-10): Each of these questions is based on the following information:
(i) P % Q means P is the father of B.
(ii) P @ Q means P is the sister of B.
(iii) P $ Q means P is the brother of B.
(iv) P * Q means P is the son of B.

Q10. In the expression N @ M * A $ S % Z, how is S related to N?
```

Option Set:
- Aunt
- Uncle
- Sister
- Sister in law
- None of these

Exact Answer: `Uncle`

Complete Mathematical / Logical Deduction:

Official solution from Blood Relation Topic Mock Test 01. Symmetrical, rotational, or deductive proof confirms answer Uncle.

Reference High-Resolution Figure / Slide:


SEC01_TEST_BLOOD _Q10 Reference

Problem 16: Blood Relations & Family Trees (`SEC01_TEST_BLOOD _Q11`) ◯ [☐ PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [☐ PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

Origin Module: `4 _ Blood Relation` | **Source Document:** `BLOOD RELATION TEST 01_QUESTIONS.json`
Problem Statement:

```
Directions (11-12): Study the information carefully and answer the questions given below.
P is father of M who is brother in law of K. N is daughter in law of Z who is grandmother of L. P has only t

Q11. If K is wife of J then what could be the relation of J with respect to N?
```

Option Set:
- Brother
- Brother in law
- Uncle
- Cannot be determined
- None of these.

Exact Answer: `Brother in law`

Complete Mathematical / Logical Deduction:

Official solution from Blood Relation Topic Mock Test 01. Symmetrical, rotational, or deductive proof confirms answer Brother in law.

Reference High-Resolution Figure / Slide:



Problem 17: Blood Relations & Family Trees (`SEC01_TEST_BLOOD _Q12`) ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

Origin Module: `4 _ Blood Relation` | **Source Document:** `BLOOD RELATION TEST 01_QUESTIONS.json`
Problem Statement:

```
Directions (11-12): Study the information carefully and answer the questions given below.
P is father of M who is brother in law of K. N is daughter in law of Z who is grandmother of L. P has only t

Q12. What is relation of L with respect to X?
```

Option Set:
- Nephew
- Niece
- Son
- Daughter
- Cannot be determined

Exact Answer: `Cannot be determined`

Complete Mathematical / Logical Deduction:

Official solution from Blood Relation Topic Mock Test 01. Symmetrical, rotational, or deductive proof confirms answer Cannot be determined.

Reference High-Resolution Figure / Slide:



Problem 18: Blood Relations & Family Trees (`SEC01_TEST_BLOOD _Q13`) ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

Origin Module: `4 _ Blood Relation` | **Source Document:** `BLOOD RELATION TEST 01_QUESTIONS.json`
Problem Statement:

```
Directions (13): Study the following information carefully to answer the given questions
P × Q means P is father of Q
P ÷ Q means P is daughter of Q
P + Q means P is sister of Q
P − Q means P is husband of Q

Q13. In M ÷ N × O − P, How O related to M?
```

Option Set:
- Brother
- Son –in-law
- Father
- Son
- None of these

Exact Answer: `Brother`

Complete Mathematical / Logical Deduction:

Official solution from Blood Relation Topic Mock Test 01. Symmetrical, rotational, or deductive proof confirms answer Brother.

Reference High-Resolution Figure / Slide:



Problem 19: Blood Relations & Family Trees (`SEC01_TEST_BLOOD _Q14`) ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

Origin Module: `4 _ Blood Relation` | **Source Document:** `BLOOD RELATION TEST 01_QUESTIONS.json`
Problem Statement:

```
Directions (14-15): Study the information carefully and answer the questions given below.
M, N, O, P, Q and R are six family members. Q is son of P, who is not the mother of Q. N is brother of P. R

Q14. How many female members are there in the family?
```

Option Set:
- Two
- Three
- Four
- Can't be determined
- None of these

Exact Answer: `Can't be determined`

Complete Mathematical / Logical Deduction:

This is the solution

Reference High-Resolution Figure / Slide:


SEC01_TEST_BLOOD _Q14 Reference

Problem 20: Blood Relations & Family Trees (`SEC01_TEST_BLOOD _Q15`) ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

Origin Module: `4 _ Blood Relation` | **Source Document:** `BLOOD RELATION TEST 01_QUESTIONS.json`
Problem Statement:

```
Directions (14-15): Study the information carefully and answer the questions given below.
M, N, O, P, Q and R are six family members. Q is son of P, who is not the mother of Q. N is brother of P. R

Q15. How is Q related to O?
```

Option Set:
- Mother
- Father
- Sister
- Brother
- None of these

Exact Answer: `Brother`

Complete Mathematical / Logical Deduction:

This is the solution

Reference High-Resolution Figure / Slide:


SEC01_TEST_BLOOD _Q15 Reference
